

Chapter 5 Exercises

SOLUTIONS TO RICK DURRETT'S
Probability: Theory and Examples

June 12, 2026

These solutions use the local Chapter 5 LLM/Viki knowledge set in `Probability/LLM_Viki_Dataset/Chapter5_Knowledge`. The cited knowledge IDs refer to entries in `chapter5_knowledge_pieces.jsonl`.

Exercise 5.1.1

Question. Let $\xi_1, \xi_2, \dots$ be i.i.d. random variables taking values in $\{1, 2, \dots, N\}$, each with probability $1/N$. Show that

$$X_n = |\{\xi_1, \dots, \xi_n\}|$$

is a Markov chain and compute its transition probability.

Knowledge used. [durrett_ch5_markov_property_countable](#);
[durrett_ch5_general_transition_probability](#).

Solution. The state X_n is the number of distinct values seen among the first n observations. If $X_n = k$, then exactly k of the N possible values have already appeared. The next observation ξ_{n+1} is independent of the past, so

$$\mathbb{P}(X_{n+1} = k + 1 \mid X_0, \dots, X_n) = \frac{N - k}{N},$$

because there are $N - k$ unseen values. Similarly,

$$\mathbb{P}(X_{n+1} = k \mid X_0, \dots, X_n) = \frac{k}{N},$$

because drawing one of the already observed values does not change the number of distinct values. These probabilities depend on the past only through X_n , so (X_n) is a Markov chain.

If we include the natural initial state $X_0 = 0$, the state space is $\{0, 1, \dots, N\}$ and the transition probability is

$$p(k, j) = \begin{cases} k/N, & j = k, \\ (N - k)/N, & j = k + 1, \\ 0, & \text{otherwise,} \end{cases} \quad 0 \leq k \leq N,$$

with the convention that $p(N, N) = 1$.

Exercise 5.1.2

Question. Let $\xi_1, \xi_2, \dots$ be i.i.d. random variables taking values in $\{-1, 1\}$, each with probability $1/2$. Let $S_0 = 0$, $S_n = \xi_1 + \dots + \xi_n$, and

$$X_n = \max\{S_m : 0 \leq m \leq n\}.$$

Show that X_n is not a Markov chain.

Knowledge used. [durrett_ch5_markov_property_countable](#); [durrett_ch5_random_walk_chain](#).

Solution. The record maximum X_n does not remember the current location S_n . That missing information affects whether the next step can create a new maximum.

Consider the process through time 3. On the event

$$A = \{X_0 = 0, X_1 = 0, X_2 = 0, X_3 = 1\},$$

we must have the increment sequence $(-1, +1, +1)$, so $S_3 = 1$ and $X_3 = 1$. Therefore

$$\mathbb{P}(X_4 = 2 \mid A) = \mathbb{P}(\xi_4 = 1) = \frac{1}{2}.$$

Now consider the event

$$B = \{X_0 = 0, X_1 = 1, X_2 = 1, X_3 = 1\}.$$

Then $\xi_1 = 1$, $\xi_2 = -1$, and ξ_3 may be either 1 or -1 . Conditional on B , these two possibilities are equally likely. If $\xi_3 = 1$, then $S_3 = 1$ and the next maximum reaches 2 with probability $1/2$. If $\xi_3 = -1$, then $S_3 = -1$ and one step cannot reach level 2. Hence

$$\mathbb{P}(X_4 = 2 \mid B) = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot 0 = \frac{1}{4}.$$

Both events have $X_3 = 1$, but the conditional probabilities of $X_4 = 2$ are different. Thus the conditional law of the future depends on more than the current value of X_n , and (X_n) is not a Markov chain.

Exercise 5.1.3

Question. Let $\xi_0, \xi_1, \dots$ be i.i.d. random variables taking values in $\{H, T\}$, each with probability $1/2$. Show that $X_n = (\xi_n, \xi_{n+1})$ is a Markov chain and compute its transition probability p . What is p^2 ?

Knowledge used. [durrett_ch5_markov_property_countable](#); [durrett_ch5_chapman_kolmogorov](#).

Solution. The state space is

$$\mathcal{S} = \{HH, HT, TH, TT\}.$$

If $X_n = (a, b)$, then

$$X_{n+1} = (\xi_{n+1}, \xi_{n+2}) = (b, \xi_{n+2}).$$

The variable ξ_{n+2} is independent of the past and is equally likely to be H or T . Therefore

$$p((a, b), (c, d)) = \begin{cases} 1/2, & c = b, \\ 0, & c \neq b. \end{cases}$$

Equivalently, from (a, b) the chain moves to (b, H) and (b, T) with probability $1/2$ each.

After two steps,

$$X_{n+2} = (\xi_{n+2}, \xi_{n+3}),$$

and ξ_{n+2} and ξ_{n+3} are independent fair coin tosses, independent of X_n . Hence

$$p^2((a, b), (c, d)) = \frac{1}{4} \quad \text{for all } (a, b), (c, d) \in \mathcal{S}.$$

In the order (HH, HT, TH, TT) ,

$$p = \begin{pmatrix} 1/2 & 1/2 & 0 & 0 \\ 0 & 0 & 1/2 & 1/2 \\ 1/2 & 1/2 & 0 & 0 \\ 0 & 0 & 1/2 & 1/2 \end{pmatrix}, \quad p^2 = \begin{pmatrix} 1/4 & 1/4 & 1/4 & 1/4 \\ 1/4 & 1/4 & 1/4 & 1/4 \\ 1/4 & 1/4 & 1/4 & 1/4 \\ 1/4 & 1/4 & 1/4 & 1/4 \end{pmatrix}.$$

Exercise 5.1.4

Question. Brother-sister mating. In this scheme, two animals are mated, and among their direct descendants two individuals of opposite sex are selected at random. These animals are mated and the process continues. Suppose each individual can be one of three genotypes AA , Aa , aa , and suppose that the type of the offspring is determined by selecting a letter from each parent. With these rules, the pair of genotypes in the n th generation is a Markov chain with six states:

$$AA, AA; \quad AA, Aa; \quad AA, aa; \quad Aa, Aa; \quad Aa, aa; \quad aa, aa.$$

Compute its transition probability.

Knowledge used. `durrett_ch5_markov_property_countable;`
`durrett_ch5_general_transition_probability.`

Solution. For a fixed parental pair, let

$$(p, q, r) = (\mathbb{P}(\text{offspring is } AA), \mathbb{P}(\text{offspring is } Aa), \mathbb{P}(\text{offspring is } aa)).$$

The next brother-sister pair consists of two independent offspring, recorded as an unordered genotype pair. Therefore the transition probabilities to the six states

$$AA, AA; \quad AA, Aa; \quad AA, aa; \quad Aa, Aa; \quad Aa, aa; \quad aa, aa$$

are

$$p^2, \quad 2pq, \quad 2pr, \quad q^2, \quad 2qr, \quad r^2.$$

The Mendelian offspring distributions for the six parental states are

parental state	(p, q, r)
AA, AA	$(1, 0, 0)$
AA, Aa	$(1/2, 1/2, 0)$
AA, aa	$(0, 1, 0)$
Aa, Aa	$(1/4, 1/2, 1/4)$
Aa, aa	$(0, 1/2, 1/2)$
aa, aa	$(0, 0, 1)$

Thus, in the state order displayed above, the transition matrix is

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1/4 & 1/2 & 0 & 1/4 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 1/16 & 1/4 & 1/8 & 1/4 & 1/4 & 1/16 \\ 0 & 0 & 0 & 1/4 & 1/2 & 1/4 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Exercise 5.1.5

Question. Bernoulli-Laplace model of diffusion. Suppose two urns, which we call left and right, have m balls each. There are b black balls, with $b \leq m$, and $2m - b$ white balls. At each time, we pick one ball from each urn and interchange them. Let the state at time n be the number of black balls in the left urn. Compute the transition probability.

Knowledge used. [durrett_ch5_markov_property_countable](#); [durrett_ch5_ehrenfest_chain](#); [durrett_ch5_general_transition_probability](#).

Solution. Since $b \leq m$, the possible states are $0, 1, \dots, b$. If the current state is i , then the left urn contains i black and $m - i$ white balls. The right urn contains $b - i$ black and $m - b + i$ white balls.

The number of black balls in the left urn increases by 1 exactly when we choose a white ball from the left urn and a black ball from the right urn. Hence

$$p(i, i + 1) = \frac{m - i}{m} \cdot \frac{b - i}{m}.$$

It decreases by 1 exactly when we choose a black ball from the left urn and a white ball from the right urn. Hence

$$p(i, i - 1) = \frac{i}{m} \cdot \frac{m - b + i}{m}.$$

It stays fixed when the two selected balls have the same color:

$$p(i, i) = \frac{i}{m} \cdot \frac{b - i}{m} + \frac{m - i}{m} \cdot \frac{m - b + i}{m}.$$

All other transition probabilities are zero. The boundary cases are included automatically because the relevant factors vanish when $i = 0$ or $i = b$.

Exercise 5.1.6

Question. Let $\theta, U_1, U_2, \dots$ be independent and uniform on $(0, 1)$. Let $X_i = 1$ if $U_i \leq \theta$ and $X_i = -1$ if $U_i > \theta$, and let $S_n = X_1 + \dots + X_n$. In words, we first pick θ according to the uniform distribution and then flip a coin with probability θ of heads to generate a random walk. Compute

$$\mathbb{P}(X_{n+1} = 1 \mid X_1, \dots, X_n)$$

and conclude that S_n is a temporally inhomogeneous Markov chain. This is due to the fact that “ S_n is a sufficient statistic for estimating θ .”

Knowledge used. [durrett_ch5_markov_property_countable](#); [durrett_ch5_random_walk_chain](#).

Solution. Let

$$H_n = \#\{1 \leq i \leq n : X_i = 1\}.$$

Then $H_n = (n + S_n)/2$. Conditional on θ , the variables $X_1, \dots, X_n$ are i.i.d. with

$$\mathbb{P}(X_i = 1 \mid \theta) = \theta.$$

Since the prior distribution of θ is uniform on $(0, 1)$, the posterior density of θ given $X_1, \dots, X_n$ is proportional to

$$\theta^{H_n} (1 - \theta)^{n - H_n}.$$

Thus the posterior distribution is

$$\theta \mid X_1, \dots, X_n \sim \text{Beta}(H_n + 1, n - H_n + 1).$$

Therefore the predictive probability of a head on the next toss is the posterior mean:

$$\mathbb{P}(X_{n+1} = 1 \mid X_1, \dots, X_n) = \mathbb{E}(\theta \mid X_1, \dots, X_n) = \frac{H_n + 1}{n + 2}.$$

In terms of S_n ,

$$\mathbb{P}(X_{n+1} = 1 \mid X_1, \dots, X_n) = \frac{(n + S_n)/2 + 1}{n + 2} = \frac{n + S_n + 2}{2n + 4}.$$

Consequently,

$$\mathbb{P}(S_{n+1} = s + 1 \mid S_0, \dots, S_n = s) = \frac{(n + s)/2 + 1}{n + 2},$$

and

$$\mathbb{P}(S_{n+1} = s - 1 \mid S_0, \dots, S_n = s) = 1 - \frac{(n + s)/2 + 1}{n + 2} = \frac{(n - s)/2 + 1}{n + 2}.$$

These transition probabilities depend only on the current value s and the time n , not on the earlier path. Hence (S_n) is a temporally inhomogeneous Markov chain.

Exercise 5.2.1

Question. Let $A \in \sigma(X_0, \dots, X_n)$ and $B \in \sigma(X_n, X_{n+1}, \dots)$. Use the Markov property to show that for any initial distribution μ ,

$$\mathbb{P}_\mu(A \cap B \mid X_n) = \mathbb{P}_\mu(A \mid X_n) \mathbb{P}_\mu(B \mid X_n).$$

In words, the past and future are conditionally independent given the present. Hint: Write the left-hand side as $\mathbb{E}_\mu(\mathbb{E}_\mu(\mathbf{1}_A \mathbf{1}_B \mid \mathcal{F}_n) \mid X_n)$.

Knowledge used. [durrett_ch5_markov_property_shift](#);
[durrett_ch5_general_transition_probability](#).

Solution. Since $A \in \mathcal{F}_n$, the tower property gives

$$\mathbb{P}_\mu(A \cap B \mid X_n) = \mathbb{E}_\mu(\mathbb{E}_\mu(\mathbf{1}_A \mathbf{1}_B \mid \mathcal{F}_n) \mid X_n) = \mathbb{E}_\mu(\mathbf{1}_A \mathbb{E}_\mu(\mathbf{1}_B \mid \mathcal{F}_n) \mid X_n).$$

The event B is a measurable function of the future path $(X_n, X_{n+1}, \dots)$. By the Markov property, there is a measurable function φ such that

$$\mathbb{E}_\mu(\mathbf{1}_B \mid \mathcal{F}_n) = \varphi(X_n).$$

Therefore

$$\mathbb{P}_\mu(A \cap B \mid X_n) = \mathbb{E}_\mu(\mathbf{1}_A \varphi(X_n) \mid X_n) = \varphi(X_n) \mathbb{E}_\mu(\mathbf{1}_A \mid X_n).$$

Also,

$$\mathbb{P}_\mu(B \mid X_n) = \mathbb{E}_\mu(\mathbb{E}_\mu(\mathbf{1}_B \mid \mathcal{F}_n) \mid X_n) = \varphi(X_n).$$

Combining the last two displays gives

$$\mathbb{P}_\mu(A \cap B \mid X_n) = \mathbb{P}_\mu(A \mid X_n) \mathbb{P}_\mu(B \mid X_n).$$

Exercise 5.2.2

Question. Let X_n be a Markov chain. Use Levy's 0-1 law to show that if

$$\mathbb{P}\left(\bigcup_{m=n+1}^{\infty} \{X_m \in B_m\} \middle| X_n\right) \geq \delta > 0 \quad \text{on } \{X_n \in A_n\},$$

then

$$\mathbb{P}(\{X_n \in A_n \text{ i.o.}\} - \{X_n \in B_n \text{ i.o.}\}) = 0.$$

Knowledge used. [durrett_ch5_markov_property_shift](#);
[durrett_ch5_hitting_return_decomposition](#).

Solution. For $N \geq 0$, set

$$D_N = \bigcup_{m=N+1}^{\infty} \{X_m \in B_m\}.$$

If $n > N$, then

$$D_N \supseteq \bigcup_{m=n+1}^{\infty} \{X_m \in B_m\}.$$

Hence, using the Markov property and the hypothesis,

$$\mathbb{P}(D_N \mid \mathcal{F}_n) \geq \mathbb{P}\left(\bigcup_{m=n+1}^{\infty} \{X_m \in B_m\} \middle| \mathcal{F}_n\right) = \mathbb{P}\left(\bigcup_{m=n+1}^{\infty} \{X_m \in B_m\} \middle| X_n\right) \geq \delta$$

on $\{X_n \in A_n\}$.

By Levy's 0-1 law,

$$\mathbb{P}(D_N \mid \mathcal{F}_n) \longrightarrow \mathbf{1}_{D_N} \quad \text{a.s.}$$

If $\{X_n \in A_n \text{ i.o.}\}$ occurs, then along infinitely many $n > N$ the conditional probability above is at least δ . Therefore $\mathbf{1}_{D_N} = 1$ on this event. Since this holds for every N ,

$$\{X_n \in A_n \text{ i.o.}\} \subseteq \bigcap_{N=0}^{\infty} D_N = \{X_n \in B_n \text{ i.o.}\}$$

up to a null set. This proves the desired result.

Exercise 5.2.3

Question. A state a is called absorbing if $P_a(X_1 = a) = 1$. Let

$$D = \{X_n = a \text{ for some } n \geq 1\}$$

and let $h(x) = P_x(D)$. Use the result of the previous exercise to conclude that $h(X_n) \rightarrow 0$ a.s. on D^c . Here a.s. means P_μ -a.s. for any initial distribution μ .

Knowledge used. [durrett_ch5_markov_property_countable](#);
[durrett_ch5_hitting_return_decomposition](#); [durrett_ch5_strong_markov_property](#).

Solution. Fix $\varepsilon > 0$ and define

$$A_n = \{x : h(x) \geq \varepsilon\}, \quad B_n = \{a\}.$$

If $X_n = x$ and $h(x) \geq \varepsilon$, then the probability of hitting a at some future positive time is at least ε :

$$\mathbb{P}\left(\bigcup_{m=n+1}^{\infty} \{X_m = a\} \mid X_n = x\right) = h(x) \geq \varepsilon.$$

Since a is absorbing, hitting a once implies $X_m = a$ for infinitely many m . Exercise 5.2.2 gives

$$\mathbb{P}(\{h(X_n) \geq \varepsilon \text{ i.o.}\} - \{X_n = a \text{ i.o.}\}) = 0.$$

On D^c , the chain never hits a at any positive time, so $\{X_n = a \text{ i.o.}\}$ does not occur. Hence

$$\mathbb{P}(\{h(X_n) \geq \varepsilon \text{ i.o.}\} \cap D^c) = 0.$$

Taking ε through the positive rationals shows that $\limsup_n h(X_n) = 0$ on D^c . Since $h \geq 0$, this is exactly $h(X_n) \rightarrow 0$ a.s. on D^c .

Exercise 5.2.4

Question. First entrance decomposition. Let $T_y = \inf\{n \geq 1 : X_n = y\}$. Show that

$$p^n(x, y) = \sum_{m=1}^n P_x(T_y = m) p^{n-m}(y, y).$$

Knowledge used. [durrett_ch5_strong_markov_property](#);
[durrett_ch5_hitting_return_decomposition](#); [durrett_ch5_chapman_kolmogorov](#).

Solution. On the event $\{X_n = y\}$, the first positive entrance time into y must be some $m \in \{1, \dots, n\}$. These events are disjoint, so

$$p^n(x, y) = P_x(X_n = y) = \sum_{m=1}^n P_x(T_y = m, X_n = y).$$

By the strong Markov property at T_y , conditional on $T_y = m$ the chain restarts from y at time m . Thus

$$P_x(X_n = y \mid T_y = m) = P_y(X_{n-m} = y) = p^{n-m}(y, y),$$

and therefore

$$p^n(x, y) = \sum_{m=1}^n P_x(T_y = m) p^{n-m}(y, y).$$

Exercise 5.2.5

Question. Show that

$$\sum_{m=0}^n P_x(X_m = x) \geq \sum_{m=k}^{n+k} P_x(X_m = x).$$

Knowledge used. [durrett_ch5_chapman_kolmogorov](#);
[durrett_ch5_hitting_return_decomposition](#).

Solution. For any state y , the expected number of visits to x during the first n steps starting from y is at most the corresponding quantity starting from x . Indeed, let $T_x = \inf\{m \geq 0 : X_m = x\}$. If $y = x$, there is equality. If $y \neq x$, then by decomposing at the first visit to x ,

$$\sum_{r=0}^n P_y(X_r = x) = \sum_{\ell=1}^n P_y(T_x = \ell) \sum_{s=0}^{n-\ell} P_x(X_s = x) \leq \sum_{s=0}^n P_x(X_s = x).$$

Now use the Markov property at time k :

$$\begin{aligned} \sum_{m=k}^{n+k} P_x(X_m = x) &= \sum_{r=0}^n P_x(X_{k+r} = x) \\ &= \sum_y P_x(X_k = y) \sum_{r=0}^n P_y(X_r = x) \\ &\leq \sum_y P_x(X_k = y) \sum_{r=0}^n P_x(X_r = x) \\ &= \sum_{r=0}^n P_x(X_r = x). \end{aligned}$$

Exercise 5.2.6

Question. Let $T_C = \inf\{n \geq 1 : X_n \in C\}$. Suppose that $S - C$ is finite and for each $x \in S - C$,

$$P_x(T_C < \infty) > 0.$$

Then there are $N < \infty$ and $\epsilon > 0$ such that for all $y \in S - C$,

$$P_y(T_C > kN) \leq (1 - \epsilon)^k.$$

Knowledge used. [durrett_ch5_strong_markov_property](#);
[durrett_ch5_hitting_return_decomposition](#).

Solution. For each $y \in S - C$, the assumption $P_y(T_C < \infty) > 0$ implies that there is an integer N_y with

$$P_y(T_C \leq N_y) > 0.$$

Since $S - C$ is finite, choose

$$N = \max_{y \in S-C} N_y, \quad \epsilon = \min_{y \in S-C} P_y(T_C \leq N).$$

Then $N < \infty$ and $\epsilon > 0$.

For $y \in S - C$,

$$P_y(T_C > N) \leq 1 - \epsilon.$$

Using the Markov property at times $N, 2N, \dots$, on the event $\{T_C > jN\}$ the chain is still in $S - C$, so the conditional probability of avoiding C for the next N steps is at most $1 - \epsilon$. Thus

$$P_y(T_C > (j+1)N \mid \mathcal{F}_{jN}) \leq 1 - \epsilon \quad \text{on } \{T_C > jN\}.$$

Induction gives

$$P_y(T_C > kN) \leq (1 - \epsilon)^k.$$

Exercise 5.2.7

Question. Exit distributions. Let $V_C = \inf\{n \geq 0 : X_n \in C\}$ and let

$$h(x) = P_x(V_A < V_B).$$

Suppose $A \cap B = \emptyset$, $S - (A \cup B)$ is finite, and $P_x(V_{A \cup B} < \infty) > 0$ for all $x \in S - (A \cup B)$.

(i) Show that

$$h(x) = \sum_y p(x, y)h(y) \quad x \notin A \cup B.$$

(ii) Show that if h satisfies this equation, then $h(X_{n \wedge V_{A \cup B}})$ is a martingale.

(iii) Use this and Exercise 5.2.6 to conclude that $h(x) = P_x(V_A < V_B)$ is the only solution of this equation that is 1 on A and 0 on B .

Knowledge used. [durrett_ch5_expectation_operator_notation](#); [durrett_ch5_hitting_return_decomposition](#); [durrett_ch5_strong_markov_property](#).

Solution. Let $\tau = V_{A \cup B}$. If $x \notin A \cup B$, then after one step the chain is at y with probability $p(x, y)$, and the future exit probability is $h(y)$. Therefore

$$h(x) = \sum_y p(x, y)h(y), \quad x \notin A \cup B.$$

Also $h = 1$ on A and $h = 0$ on B .

Now suppose h is any function satisfying the displayed equation off $A \cup B$. If $n \geq \tau$, then $h(X_{(n+1) \wedge \tau}) = h(X_{n \wedge \tau})$. If $n < \tau$, then $X_n \notin A \cup B$, so

$$\mathbb{E}(h(X_{(n+1) \wedge \tau}) \mid \mathcal{F}_n) = \sum_y p(X_n, y)h(y) = h(X_n) = h(X_{n \wedge \tau}).$$

Thus $h(X_{n \wedge \tau})$ is a martingale.

By Exercise 5.2.6, $\tau < \infty$ a.s. and in fact its tail decays geometrically. Since $S - (A \cup B)$ is finite and the boundary values are fixed, any solution h is bounded. Taking expectations in the stopped martingale identity gives

$$h(x) = E_x h(X_{n \wedge \tau}).$$

Letting $n \rightarrow \infty$ and using bounded convergence,

$$h(x) = E_x h(X_\tau) = P_x(X_\tau \in A) = P_x(V_A < V_B).$$

Therefore the exit probability is the unique solution with boundary values 1 on A and 0 on B .

Exercise 5.2.8

Question. Let X_n be a Markov chain with $S = \{0, 1, \dots, N\}$ and suppose that X_n is a martingale. Let $V_x = \min\{n \geq 0 : X_n = x\}$ and suppose $P_x(V_0 \wedge V_N < \infty) > 0$ for all x . Show that

$$P_x(V_N < V_0) = \frac{x}{N}.$$

Knowledge used. [durrett_ch5_expectation_operator_notation](#);
[durrett_ch5_hitting_return_decomposition](#).

Solution. Let

$$h(x) = P_x(V_N < V_0).$$

By Exercise 5.2.7, h is the unique function that is harmonic on $\{1, \dots, N-1\}$ and satisfies $h(0) = 0$, $h(N) = 1$.

Since (X_n) is a martingale,

$$E_x X_1 = x.$$

Hence the function $f(x) = x/N$ satisfies

$$\sum_y p(x, y) f(y) = \frac{1}{N} \sum_y p(x, y) y = \frac{x}{N} = f(x), \quad 1 \leq x \leq N-1.$$

It also has boundary values $f(0) = 0$ and $f(N) = 1$. By uniqueness,

$$h(x) = f(x) = \frac{x}{N}.$$

Exercise 5.2.9

Question. Wright-Fisher model. Suppose $S = \{0, 1, \dots, N\}$ and consider

$$p(i, j) = \binom{N}{j} \left(\frac{i}{N}\right)^j \left(1 - \frac{i}{N}\right)^{N-j}.$$

Show that this chain satisfies the hypotheses of Exercise 5.2.8.

Knowledge used. [durrett_ch5_wright_fisher_chain](#); [durrett_ch5_markov_property_countable](#).

Solution. Given $X_n = i$, the next state has distribution

$$X_{n+1} \sim \text{Binomial}\left(N, \frac{i}{N}\right).$$

Therefore

$$E(X_{n+1} \mid X_n = i) = N \cdot \frac{i}{N} = i.$$

Thus (X_n) is a martingale.

The state space is $\{0, 1, \dots, N\}$. Also 0 and N are absorbing: if $i = 0$, then $X_{n+1} = 0$ a.s.; if $i = N$, then $X_{n+1} = N$ a.s. For $1 \leq i \leq N-1$,

$$p(i, 0) = \left(1 - \frac{i}{N}\right)^N > 0, \quad p(i, N) = \left(\frac{i}{N}\right)^N > 0.$$

Hence from every interior state there is positive probability of hitting 0 or N in one step. Thus $P_i(V_0 \wedge V_N < \infty) > 0$ for all i , and all hypotheses of Exercise 5.2.8 are satisfied.

Exercise 5.2.10

Question. In brother-sister mating described in Exercise 5.1.4, AA, AA and aa, aa are absorbing states. Show that the number of A 's in the pair is a martingale and use this to compute the probability of getting absorbed in AA, AA starting from each of the states.

Knowledge used. [durrett_ch5_markov_property_countable](#);
[durrett_ch5_hitting_return_decomposition](#); [durrett_ch5_general_transition_probability](#).

Solution. Let M_n be the number of A alleles in the genotype pair at generation n . Thus M_n takes values in $\{0, 1, 2, 3, 4\}$. Given a parental pair with K total A alleles, one offspring receives one randomly chosen allele from each parent. If the first parent has k_1 A alleles and the second has k_2 A alleles, then $K = k_1 + k_2$ and the expected number of A alleles in one offspring is

$$\frac{k_1}{2} + \frac{k_2}{2} = \frac{K}{2}.$$

The next brother-sister pair consists of two offspring, so

$$E(M_{n+1} \mid \mathcal{F}_n) = 2 \cdot \frac{M_n}{2} = M_n.$$

Thus (M_n) is a martingale.

The only absorbing states are AA, AA and aa, aa . From each nonabsorbing state there is positive probability of eventually reaching one of them, and the state space is finite, so the absorption time is finite a.s. If

$$q = \mathbb{P}(\text{absorption in } AA, AA),$$

then at absorption the martingale has value 4 with probability q and 0 with probability $1 - q$. Optional stopping gives

$$M_0 = E(M_T) = 4q, \quad \text{so} \quad q = \frac{M_0}{4}.$$

The absorption probabilities in the state order

$$AA, AA; \quad AA, Aa; \quad AA, aa; \quad Aa, Aa; \quad Aa, aa; \quad aa, aa$$

are therefore

$$1, \quad \frac{3}{4}, \quad \frac{1}{2}, \quad \frac{1}{2}, \quad \frac{1}{4}, \quad 0.$$

Exercise 5.2.11

Question. Exit times. Let $V_A = \inf\{n \geq 0 : X_n \in A\}$ and $g(x) = E_x V_A$. Suppose that $S - A$ is finite and for each $x \in S - A$, $P_x(V_A < \infty) > 0$.

(i) Show that

$$g(x) = 1 + \sum_y p(x, y)g(y) \quad x \notin A.$$

(ii) Show that if g satisfies this equation, then $g(X_{n \wedge V_A}) + n \wedge V_A$ is a martingale.

(iii) Use this to conclude that $g(x) = E_x V_A$ is the only solution of this equation that is 0 on A .

Knowledge used. [durrett_ch5_expectation_operator_notation](#);
[durrett_ch5_hitting_return_decomposition](#); [durrett_ch5_strong_markov_property](#).

Solution. Let $\tau = V_A$. Exercise 5.2.6 implies that $\tau < \infty$ a.s. and has a geometrically decaying tail, so $E_x \tau < \infty$ for all x .

If $x \notin A$, then after one step the chain moves to y with probability $p(x, y)$. One unit of time has elapsed, and the remaining expected time is $g(y)$. Thus

$$g(x) = 1 + \sum_y p(x, y)g(y), \quad x \notin A.$$

Also $g(x) = 0$ for $x \in A$.

Now suppose g satisfies this equation and is 0 on A . Let

$$M_n = g(X_{n \wedge \tau}) + n \wedge \tau.$$

If $n \geq \tau$, then $M_{n+1} = M_n$. If $n < \tau$, then $X_n \notin A$ and

$$\begin{aligned} E(M_{n+1} \mid \mathcal{F}_n) &= E(g(X_{n+1}) + n + 1 \mid \mathcal{F}_n) \\ &= \sum_y p(X_n, y)g(y) + n + 1 \\ &= g(X_n) + n = M_n. \end{aligned}$$

Therefore (M_n) is a martingale.

Taking expectations gives

$$g(x) = E_x[g(X_{n \wedge \tau}) + n \wedge \tau].$$

Letting $n \rightarrow \infty$, we have $X_\tau \in A$, so $g(X_\tau) = 0$, and $n \wedge \tau \rightarrow \tau$. Hence

$$g(x) = E_x \tau = E_x V_A.$$

Thus the expected hitting time is the unique solution with boundary value 0 on A .

Exercise 5.2.12

Question. In this problem we imagine that you are bored and start flipping a coin to see how many tosses $N(HH)$ you need until you get two heads in a row. Let $\xi_0, \xi_1, \dots$ be i.i.d. random variables taking values in $\{H, T\}$, each with probability $1/2$, and let $X_n = (\xi_n, \xi_{n+1})$ be the Markov chain from Exercise 5.1.3. Use the result of the previous exercise to compute

$$g(x) = \text{the expected time to reach state } (H, H) \text{ starting from } x.$$

The answer we want is

$$EN(HH) = 2 + \frac{1}{4}[g(H, H) + g(H, T) + g(T, H) + g(T, T)].$$

Knowledge used. [durrett_ch5_markov_property_countable](#); [durrett_ch5_chapman_kolmogorov](#);
[durrett_ch5_expectation_operator_notation](#).

Solution. Let the target state be HH . Then $g(HH) = 0$. From Exercise 5.1.3, the transitions are

$$(a, b) \longrightarrow (b, H) \text{ or } (b, T), \quad \text{each with probability } 1/2.$$

Write

$$a = g(HT), \quad b = g(TH), \quad c = g(TT).$$

Using Exercise 5.2.11, the equations are

$$a = 1 + \frac{1}{2}b + \frac{1}{2}c,$$

$$b = 1 + \frac{1}{2}g(HH) + \frac{1}{2}a = 1 + \frac{1}{2}a,$$

and

$$c = 1 + \frac{1}{2}b + \frac{1}{2}c.$$

The first and third equations imply $a = c$. Hence

$$a = 1 + \frac{1}{2}b + \frac{1}{2}a, \quad \text{so} \quad a = 2 + b.$$

Together with $b = 1 + a/2$, this gives

$$a = 6, \quad b = 4, \quad c = 6.$$

Thus

$$g(HH) = 0, \quad g(HT) = 6, \quad g(TH) = 4, \quad g(TT) = 6.$$

The first two tosses create one of the four pair states uniformly, so

$$EN(HH) = 2 + \frac{1}{4}(0 + 6 + 4 + 6) = 6.$$

Exercise 5.2.13

Question. In this problem we consider the number of tosses $N(HHH)$ to get HHH . Let X_n be the number of consecutive heads we have after n tosses. Then X_n is a Markov chain with transition probability

	0	1	2	3
0	1/2	1/2	0	0
1	1/2	0	1/2	0
2	1/2	0	0	1/2
3	0	0	0	1

Let T_3 be the time to reach state 3. Use Problem 5.2.11 to compute E_0T_3 .

Knowledge used. [durrett_ch5_markov_property_countable](#);
[durrett_ch5_expectation_operator_notation](#); [durrett_ch5_hitting_return_decomposition](#).

Solution. Let

$$g_i = E_iT_3, \quad i = 0, 1, 2, 3.$$

Then $g_3 = 0$. By Exercise 5.2.11, for $i = 0, 1, 2$,

$$g_i = 1 + \sum_j p(i, j)g_j.$$

Using the displayed transition matrix,

$$g_0 = 1 + \frac{1}{2}g_0 + \frac{1}{2}g_1,$$

$$g_1 = 1 + \frac{1}{2}g_0 + \frac{1}{2}g_2,$$

and

$$g_2 = 1 + \frac{1}{2}g_0 + \frac{1}{2}g_3 = 1 + \frac{1}{2}g_0.$$

Substitute the last equation into the equation for g_1 :

$$g_1 = 1 + \frac{1}{2}g_0 + \frac{1}{2} \left(1 + \frac{1}{2}g_0 \right) = \frac{3}{2} + \frac{3}{4}g_0.$$

The equation for g_0 gives $g_0 = 2 + g_1$, so

$$g_0 = 2 + \frac{3}{2} + \frac{3}{4}g_0 = \frac{7}{2} + \frac{3}{4}g_0.$$

Therefore

$$\frac{1}{4}g_0 = \frac{7}{2}, \quad \text{and hence} \quad E_0T_3 = g_0 = 14.$$

Exercise 5.3.1

Question. Suppose y is recurrent and, for $k \geq 0$, let $R_k = T_y^k$ be the time of the k th return to y . For $k \geq 1$, let $r_k = R_k - R_{k-1}$ be the k th interarrival time. Use the strong Markov property to conclude that under P_y , the excursion vectors

$$v_k = (r_k, X_{R_{k-1}}, X_{R_{k-1}+1}, \dots, X_{R_k-1}), \quad k \geq 1,$$

are i.i.d.

Knowledge used. [durrett_ch5_strong_markov_property](#);
[durrett_ch5_hitting_return_decomposition](#); [durrett_ch5_recurrence_visit_criterion](#).

Solution. Since y is recurrent, under P_y all return times R_k are finite almost surely. The time R_{k-1} is a stopping time and $X_{R_{k-1}} = y$. By the strong Markov property, conditional on $\mathcal{F}_{R_{k-1}}$, the shifted process

$$(X_{R_{k-1}+m})_{m \geq 0}$$

has the same law as a fresh chain started from y , and is independent of the past before R_{k-1} .

Therefore the path segment from R_{k-1} up to, but not including, the next return to y has the same distribution for every k . Moreover, each such segment is conditionally independent of all earlier segments. Thus the excursion vectors $v_1, v_2, \dots$ are independent and identically distributed under P_y .

Exercise 5.3.2

Question. Use the strong Markov property to show that

$$\rho_{xz} \geq \rho_{xy}\rho_{yz}.$$

Knowledge used. [durrett_ch5_strong_markov_property](#);
[durrett_ch5_hitting_return_decomposition](#).

Solution. Let $T_y = \inf\{n \geq 1 : X_n = y\}$ and $T_z = \inf\{n \geq 1 : X_n = z\}$. If the chain first hits y and then, from y , eventually hits z , then it has hit z starting from x . By the strong Markov property at T_y ,

$$\begin{aligned}\rho_{xz} &= P_x(T_z < \infty) \\ &\geq P_x(T_y < \infty, T_z \circ \theta_{T_y} < \infty) \\ &= E_x[\mathbf{1}_{\{T_y < \infty\}} P_y(T_z < \infty)] \\ &= \rho_{xy} \rho_{yz}.\end{aligned}$$

Exercise 5.3.3

Question. Show that in the Ehrenfest chain from Example 5.1.3 all states are recurrent.

Knowledge used. `durrett_ch5_ehrenfest_chain`; `durrett_ch5_finite_closed_recurrent_class`; `durrett_ch5_communication_recurrence_transfer`.

Solution. The Ehrenfest chain has finite state space $\{0, 1, \dots, r\}$ and transition probabilities

$$p(k, k+1) = \frac{r-k}{r}, \quad p(k, k-1) = \frac{k}{r}.$$

From any state i one can reach any state j by moving one step at a time upward or downward, and each required step has positive probability. Hence the whole finite state space is irreducible. It is also closed, since the chain never leaves $\{0, \dots, r\}$.

A finite closed irreducible class consists entirely of recurrent states. Therefore every state of the Ehrenfest chain is recurrent.

Exercise 5.3.4

Question. To probe the boundary between recurrence and transience for birth and death chains, suppose

$$p_j = \frac{1}{2} + \epsilon_j, \quad \epsilon_j \sim Cj^{-\alpha} \quad \text{as } j \rightarrow \infty, \quad q_j = 1 - p_j.$$

Show that:

- (i) if $\alpha > 1$, then 0 is recurrent;
- (ii) if $\alpha < 1$, then 0 is transient;
- (iii) if $\alpha = 1$, then 0 is transient if $C > 1/4$ and recurrent if $C < 1/4$.

Knowledge used. `durrett_ch5_birth_death_scale_function`; `durrett_ch5_birth_death_recurrence_test`.

Solution. For a birth and death chain, the recurrence test is

$$\sum_{m=0}^{\infty} \prod_{j=1}^m \frac{q_j}{p_j} = \infty.$$

Here

$$\frac{q_j}{p_j} = \frac{1/2 - \epsilon_j}{1/2 + \epsilon_j} = \frac{1 - 2\epsilon_j}{1 + 2\epsilon_j}.$$

For small ϵ_j ,

$$\log \frac{q_j}{p_j} = -4\epsilon_j + O(\epsilon_j^2).$$

Thus

$$\log \prod_{j=1}^m \frac{q_j}{p_j} = -4 \sum_{j=1}^m \epsilon_j + O\left(\sum_{j=1}^m \epsilon_j^2\right).$$

If $\alpha > 1$, then $\sum_j \epsilon_j$ converges, and so the products $\prod_{j=1}^m (q_j/p_j)$ converge to a positive finite limit. The series of products diverges, so 0 is recurrent.

If $\alpha < 1$ and $C > 0$, then

$$\sum_{j=1}^m \epsilon_j \sim \frac{C}{1-\alpha} m^{1-\alpha}.$$

The products are bounded above by

$$\exp\{-cm^{1-\alpha}\}$$

for some $c > 0$ and large m , so the series of products converges. Thus 0 is transient. This is the outward-drift case intended by the statement; if $C < 0$, the drift is eventually toward 0 and the same test gives recurrence.

Finally suppose $\alpha = 1$. Then

$$\sum_{j=1}^m \epsilon_j \sim C \log m,$$

and since $\sum_j \epsilon_j^2 < \infty$,

$$\prod_{j=1}^m \frac{q_j}{p_j} = m^{-4C+o(1)}.$$

Therefore the product series behaves like a p -series with exponent $4C$. It diverges when $4C < 1$ and converges when $4C > 1$. Hence 0 is recurrent if $C < 1/4$ and transient if $C > 1/4$.

Exercise 5.3.5

Question. Show that if we replace “ $\varphi(x) \rightarrow \infty$ ” by “ $\varphi(x) \rightarrow 0$ ” in Theorem 5.3.8 and assume that $\varphi(x) > 0$ for $x \in F$, then we can conclude that the chain is transient.

Knowledge used. [durrett_ch5_foster_supermartingale_recurrence](#); [durrett_ch5_recurrence_visit_criterion](#).

Solution. Assume the chain is irreducible and that

$$E_x \varphi(X_1) \leq \varphi(x) \quad x \notin F,$$

where F is finite, $\varphi \geq 0$, $\varphi(x) \rightarrow 0$ as $x \rightarrow \infty$, and $\varphi > 0$ on F .

Let

$$c = \min_{z \in F} \varphi(z) > 0.$$

Since $\varphi(x) \rightarrow 0$, choose $x \notin F$ with $\varphi(x) < c$. Let

$$\tau = \inf\{n \geq 0 : X_n \in F\}.$$

Then $\varphi(X_{n \wedge \tau})$ is a nonnegative supermartingale. If the chain were recurrent, irreducibility would imply $P_x(\tau < \infty) = 1$. By optional stopping and Fatou's lemma,

$$\varphi(x) \geq E_x \varphi(X_\tau) \geq c P_x(\tau < \infty) = c,$$

contradicting $\varphi(x) < c$.

Thus the irreducible chain cannot be recurrent. In an irreducible chain, all states are either recurrent or transient, so the chain is transient.

Exercise 5.3.6

Question. Let X_n be a birth and death chain with

$$p_j - \frac{1}{2} \sim \frac{C}{j} \quad \text{as } j \rightarrow \infty, \quad q_j = 1 - p_j.$$

- (i) Show that if $C < 1/4$, then we can pick $\alpha > 0$ so that $\varphi(x) = x^\alpha$ satisfies the hypotheses of Theorem 5.3.8.
- (ii) Show that when $C > 1/4$, we can take $\alpha < 0$ and apply Exercise 5.3.5.

Knowledge used. [durrett_ch5_foster_supermartingale_recurrence](#);
[durrett_ch5_birth_death_recurrence_test](#).

Solution. For large j , write $p_j = 1/2 + C/j + o(1/j)$ and $q_j = 1/2 - C/j + o(1/j)$. For $\varphi(x) = x^\alpha$, Taylor expansion gives

$$(j+1)^\alpha - j^\alpha = \alpha j^{\alpha-1} + \frac{\alpha(\alpha-1)}{2} j^{\alpha-2} + o(j^{\alpha-2}),$$

and

$$(j-1)^\alpha - j^\alpha = -\alpha j^{\alpha-1} + \frac{\alpha(\alpha-1)}{2} j^{\alpha-2} + o(j^{\alpha-2}).$$

Hence

$$\begin{aligned} E_j \varphi(X_1) - \varphi(j) &= p_j((j+1)^\alpha - j^\alpha) + q_j((j-1)^\alpha - j^\alpha) \\ &= \alpha \left(2C + \frac{\alpha-1}{2} \right) j^{\alpha-2} + o(j^{\alpha-2}). \end{aligned}$$

If $C < 1/4$, choose $\alpha > 0$ small enough that

$$2C + \frac{\alpha-1}{2} < 0, \quad \text{equivalently} \quad \alpha < 1 - 4C.$$

Then $E_j \varphi(X_1) \leq \varphi(j)$ for all sufficiently large j , and $\varphi(j) = j^\alpha \rightarrow \infty$. Taking the exceptional small states as the finite set F , Theorem 5.3.8 implies recurrence.

If $C > 1/4$, then $1 - 4C < 0$. Choose

$$1 - 4C < \alpha < 0.$$

Now $\alpha < 0$ and $2C + (\alpha-1)/2 > 0$, so their product is negative. Therefore $E_j \varphi(X_1) \leq \varphi(j)$ for all sufficiently large j , while $\varphi(j) = j^\alpha \rightarrow 0$ and $\varphi > 0$ on every finite set. Exercise 5.3.5 applies and gives transience.

Exercise 5.3.7

Question. A function f is said to be superharmonic if

$$f(x) \geq \sum_y p(x, y)f(y),$$

or equivalently $f(X_n)$ is a supermartingale. Suppose p is irreducible. Show that p is recurrent if and only if every nonnegative superharmonic function is constant.

Knowledge used. `durrett_ch5_recurrence_visit_criterion`;
`durrett_ch5_communication_recurrence_transfer`;
`durrett_ch5_foster_supermartingale_recurrence`.

Solution. First suppose the chain is recurrent and let $f \geq 0$ be superharmonic. For states x, y , irreducibility and recurrence imply $P_x(T_y < \infty) = 1$. Since $f(X_n)$ is a nonnegative supermartingale, optional stopping at $n \wedge T_y$ and Fatou's lemma give

$$f(x) \geq E_x f(X_{T_y}) = f(y).$$

Interchanging x and y gives $f(y) \geq f(x)$, so $f(x) = f(y)$. Since x, y were arbitrary, f is constant.

Conversely, suppose the irreducible chain is transient. Fix a state a and define

$$f(x) = P_x(T_a < \infty).$$

Then $f \geq 0$ and $f(a) = 1$. For $x \neq a$, first-step decomposition gives

$$f(x) = \sum_y p(x, y)f(y).$$

At $x = a$,

$$\sum_y p(a, y)f(y) = P_a(T_a < \infty) = \rho_{aa} < 1 = f(a),$$

because a is transient. Thus f is nonnegative and superharmonic. It is not constant, since $f(a) = 1$ but the displayed strict inequality would be impossible for a constant function. Therefore if every nonnegative superharmonic function is constant, the chain must be recurrent.

Exercise 5.3.8

Question. $M/M/\infty$ queue. Consider a telephone system with an infinite number of lines. Let X_n be the number of lines in use at time n , and suppose

$$X_{n+1} = \sum_{m=1}^{X_n} \xi_{n,m} + Y_{n+1},$$

where the $\xi_{n,m}$ are i.i.d. with $P(\xi_{n,m} = 1) = p$ and $P(\xi_{n,m} = 0) = 1 - p$, and Y_n is an independent i.i.d. sequence of Poisson mean λ random variables. In words, for each conversation we flip a coin with probability p of heads to see if it continues for another minute. Meanwhile, a Poisson mean λ number of conversations start between time n and $n + 1$. Use Theorem 5.3.8 with $\varphi(x) = x$ to show that the chain is recurrent for any $p < 1$.

Knowledge used. `durrett_ch5_foster_supermartingale_recurrence`;
`durrett_ch5_markov_property_countable`.

Solution. Given $X_n = x$, the number of continuing old conversations has binomial(x, p) distribution, and the number of new conversations has Poisson mean λ . Therefore

$$E_x X_1 = px + \lambda.$$

Let $\varphi(x) = x$. Since $p < 1$, choose a finite set

$$F = \{0, 1, \dots, K\}$$

with K large enough that

$$px + \lambda \leq x \quad \text{for all } x > K.$$

Equivalently, take $K \geq \lambda/(1-p)$. Then for $x \notin F$,

$$E_x \varphi(X_1) = E_x X_1 = px + \lambda \leq x = \varphi(x).$$

Also $\varphi(x) = x \rightarrow \infty$.

If $\lambda > 0$, the chain is irreducible on $\{0, 1, 2, \dots\}$: from any x all current conversations can end with positive probability, and the Poisson arrivals can produce any desired number of new conversations with positive probability. Theorem 5.3.8 therefore implies the chain is recurrent. If $\lambda = 0$, state 0 is absorbing and every initial state reaches it with probability one when $p < 1$, so the recurrent class is again evident.

Exercise 5.4.1

Question. For a random walk on $\mathbb{R}$, show that there are only four possibilities, one of which has probability one:

- (i) $S_n = 0$ for all n .
- (ii) $S_n \rightarrow \infty$.
- (iii) $S_n \rightarrow -\infty$.
- (iv) $-\infty = \liminf S_n < \limsup S_n = \infty$.

Knowledge used. [durrett_ch5_random_walk_chain](#);
[durrett_ch5_random_walk_recurrent_values](#); [durrett_ch5_random_walk_recurrence_equivalences](#).

Solution. Let

$$M^+ = \sup_{n \geq 0} S_n, \quad M^- = \sup_{n \geq 0} (-S_n) = -\inf_{n \geq 0} S_n.$$

The events $\{M^+ = \infty\}$ and $\{M^- = \infty\}$ are tail events for the iid increments, so each has probability 0 or 1.

If both have probability 1, then

$$\limsup_{n \rightarrow \infty} S_n = \infty, \quad \liminf_{n \rightarrow \infty} S_n = -\infty,$$

which is case (iv).

If $\mathbb{P}(M^+ = \infty) = 1$ and $\mathbb{P}(M^- = \infty) = 0$, then the walk is unbounded above but bounded below. The standard ladder-time argument then implies $S_n \rightarrow \infty$: if the walk returned below a fixed level infinitely often while also being unbounded above, then independent future excursions

from those return times would, with probability one, eventually create arbitrarily low new records, contradicting $M^- < \infty$. This is case (ii). By applying the same argument to $-S_n$, the case $\mathbb{P}(M^+ = \infty) = 0$ and $\mathbb{P}(M^- = \infty) = 1$ gives $S_n \rightarrow -\infty$, which is case (iii).

It remains to consider the case in which both M^+ and M^- are finite a.s. Then the random walk is a.s. bounded. If the increment law were not concentrated at 0, then for some $\varepsilon > 0$ either $\mathbb{P}(\xi_1 > \varepsilon) > 0$ or $\mathbb{P}(\xi_1 < -\varepsilon) > 0$. In the first case, arbitrarily long runs of increments larger than ε occur a.s.; in the second case, arbitrarily long runs of increments smaller than $-\varepsilon$ occur a.s. Either kind of run contradicts boundedness of the partial sums. Hence $\xi_1 = 0$ a.s., and $S_n = 0$ for all n , which is case (i).

Exercise 5.4.2

Question. Ladder variables. Let

$$\alpha(\omega) = \inf\{n : \omega_1 + \cdots + \omega_n > 0\},$$

where $\inf \emptyset = \infty$, and set $\alpha(\Delta) = \infty$. Let $\alpha_0 = 0$ and

$$\alpha_k(\omega) = \alpha_{k-1}(\omega) + \alpha(\theta_{\alpha_{k-1}}\omega), \quad k \geq 1.$$

At time α_k , the random walk is at a record high value.

- (i) If $P(\alpha < \infty) < 1$, then $P(\sup S_n < \infty) = 1$.
- (ii) If $P(\alpha < \infty) = 1$, then $P(\sup S_n = \infty) = 1$.

Knowledge used. [durrett_ch5_strong_markov_property](#); [durrett_ch5_random_walk_chain](#); [durrett_ch5_hitting_return_decomposition](#).

Solution. Let $p = P(\alpha < \infty)$. By the strong Markov property applied at the successive ladder times,

$$P(\alpha_k < \infty) = p^k.$$

Indeed, each time a new strict record is reached, the future increments are independent of the past and have the same law as the original increments.

If $p < 1$, then $P(\alpha_k < \infty) = p^k \rightarrow 0$. Hence with probability one, only finitely many strict ascending ladder times occur. After the last one, the walk never rises above the last record value, so

$$\sup_{n \geq 0} S_n < \infty \quad \text{a.s.}$$

If $p = 1$, then every α_k is finite a.s. The ladder height increments

$$S_{\alpha_k} - S_{\alpha_{k-1}}, \quad k \geq 1,$$

are iid and strictly positive. Their partial sums therefore diverge to ∞ a.s. Since S_{α_k} is the k th strict record value,

$$\sup_{n \geq 0} S_n = \infty \quad \text{a.s.}$$

Exercise 5.4.3

Question. Suppose f is superharmonic on $\mathbb{R}^d$, i.e.

$$f(x) \geq \frac{1}{|B(x, r)|} \int_{B(x, r)} f(y) dy.$$

Let $\xi_1, \xi_2, \dots$ be i.i.d. uniform on $B(0, 1)$, and define S_n by $S_n = S_{n-1} + \xi_n$ for $n \geq 1$ and $S_0 = x$.

- (i) Show that $X_n = f(S_n)$ is a supermartingale.
- (ii) Use this to conclude that in $d \leq 2$, nonnegative superharmonic functions must be constant. The example $f(x) = |x|^{2-d}$ shows this is false in $d > 2$.

Knowledge used. [durrett_ch5_chung_fuchs_one_dimension](#);
[durrett_ch5_two_dimensional_normal_recurrence](#); [durrett_ch5_random_walk_recurrent_values](#);
[durrett_ch5_foster_supermartingale_recurrence](#).

Solution. Since ξ_{n+1} is uniform on $B(0, 1)$ and independent of $\mathcal{F}_n$,

$$\begin{aligned} E(f(S_{n+1}) \mid \mathcal{F}_n) &= \frac{1}{|B(0, 1)|} \int_{B(0, 1)} f(S_n + z) dz \\ &= \frac{1}{|B(S_n, 1)|} \int_{B(S_n, 1)} f(y) dy \\ &\leq f(S_n). \end{aligned}$$

Thus $f(S_n)$ is a supermartingale whenever the expectations are well-defined; for nonnegative f , this is understood in the usual extended nonnegative sense.

Now assume $d \leq 2$ and $f \geq 0$. The increment distribution is centered and nondegenerate. In $d = 1$, Chung-Fuchs recurrence applies; in $d = 2$, the central limit theorem gives a nondegenerate normal limit after $\sqrt{n}$ scaling, so the planar recurrence theorem applies. Hence the random walk is recurrent, and because the increment law has positive density on a ball, every nonempty open ball is visited infinitely often from every starting point.

The nonnegative supermartingale $f(S_n)$ converges a.s. If two points could have different superharmonic values, recurrence would force the walk to visit neighborhoods of both points infinitely often, contradicting the existence of a single almost sure limit. Equivalently, this is the general-state analogue of the recurrent-chain fact that every nonnegative superharmonic function is constant. Therefore every nonnegative superharmonic function on $\mathbb{R}^d$ is constant for $d \leq 2$.

Exercise 5.4.4

Question. Suppose h is harmonic on $\mathbb{R}^d$, i.e.

$$h(x) = \frac{1}{|B(x, r)|} \int_{B(x, r)} h(y) dy.$$

Let $\xi_1, \xi_2, \dots$ be i.i.d. uniform on $B(0, 1)$, and define S_n by $S_n = S_{n-1} + \xi_n$ for $n \geq 1$ and $S_0 = x$.

- (i) Show that $X_n = h(S_n)$ is a martingale.
- (ii) Use this to conclude that in any dimension, bounded harmonic functions must be constant. The Hewitt-Savage 0-1 law will be useful here.

Knowledge used. [durrett_ch5_random_walk_chain](#); [durrett_ch5_markov_property_countable](#); [durrett_ch5_random_walk_recurrence_equivalences](#).

Solution. As in the previous exercise, conditioning on $\mathcal{F}_n$ gives

$$\begin{aligned} E(h(S_{n+1}) \mid \mathcal{F}_n) &= \frac{1}{|B(0, 1)|} \int_{B(0, 1)} h(S_n + z) dz \\ &= \frac{1}{|B(S_n, 1)|} \int_{B(S_n, 1)} h(y) dy \\ &= h(S_n). \end{aligned}$$

Thus $h(S_n)$ is a martingale.

Assume now that h is bounded. Then $h(S_n)$ is a bounded martingale, so it converges a.s. to a random variable L_x when $S_0 = x$. If we permute finitely many increments, then all sufficiently late partial sums are unchanged, since the total sum of those finitely many increments is unchanged. Hence L_x is invariant under finite permutations of the iid sequence (ξ_i) . By the Hewitt-Savage 0-1 law, L_x is a.s. constant. Since

$$h(x) = E_x h(S_n) = E_x L_x,$$

this constant is $h(x)$.

It remains to see that the constant does not depend on x . Fix $k \geq 1$. By the Markov property,

$$L_x = L_{S_k} \quad \text{a.s.}$$

The distribution of S_k has a density that is positive on the ball $B(x, k)$, so $h(S_k) = h(x)$ a.s. implies that $h(z) = h(x)$ for Lebesgue almost every $z \in B(x, k)$. Since k is arbitrary, h is a.e. equal to the constant $h(x)$ on all of $\mathbb{R}^d$. Applying the harmonic mean-value identity once more yields equality at every point. Therefore every bounded harmonic function on $\mathbb{R}^d$ is constant.

Exercise 5.5.1

Question. Find the stationary distribution for the Bernoulli-Laplace model of diffusion from Exercise 5.1.5.

Knowledge used. [durrett_ch5_stationary_measure_definition](#); [durrett_ch5_reversibility_detailed_balance](#); [durrett_ch5_birth_death_stationary_distribution](#).

Solution. The state i is the number of black balls in the left urn. Since $b \leq m$, the state space is $\{0, 1, \dots, b\}$. From Exercise 5.1.5,

$$p(i, i+1) = \frac{(m-i)(b-i)}{m^2}, \quad p(i, i-1) = \frac{i(m-b+i)}{m^2}.$$

At equilibrium, the left urn should look like a uniformly chosen set of m balls from the $2m$ total balls. Hence the number of black balls in the left urn has the hypergeometric distribution

$$\pi(i) = \frac{\binom{b}{i} \binom{2m-b}{m-i}}{\binom{2m}{m}}, \quad 0 \leq i \leq b.$$

We can verify this by detailed balance. For $0 \leq i < b$,

$$\frac{\pi(i+1)}{\pi(i)} = \frac{b-i}{i+1} \cdot \frac{m-i}{m-b+i+1}.$$

Also

$$\frac{p(i, i+1)}{p(i+1, i)} = \frac{(m-i)(b-i)}{(i+1)(m-b+i+1)}.$$

Thus

$$\pi(i)p(i, i+1) = \pi(i+1)p(i+1, i),$$

so π satisfies detailed balance and is stationary.

Exercise 5.5.2

Question. Let

$$w_{xy} = P_x(T_y < T_x).$$

Show that

$$\mu_x(y) = \frac{w_{xy}}{w_{yx}}.$$

Knowledge used. `durrett_ch5_stationary_measure_from_returns`;
`durrett_ch5_hitting_return_decomposition`; `durrett_ch5_strong_markov_property`.

Solution. Assume x is recurrent and y is in the same recurrent class. The quantity $\mu_x(y)$ is the expected number of visits to y before the first positive return to x , starting from x .

First, the chain must hit y before returning to x , an event with probability w_{xy} . Once it reaches y , count visits to y before hitting x . Starting from y , each visit to y starts a new trial: with probability

$$w_{yx} = P_y(T_x < T_y),$$

the chain hits x before returning again to y ; otherwise it returns to y first and the experiment restarts. Therefore the number of visits to y before hitting x , counting the initial visit to y , has geometric mean $1/w_{yx}$.

By the strong Markov property at the first hit of y ,

$$\mu_x(y) = w_{xy} \cdot \frac{1}{w_{yx}} = \frac{w_{xy}}{w_{yx}}.$$

Exercise 5.5.3

Question. Show that if p is irreducible and recurrent, then

$$\mu_x(y)\mu_y(z) = \mu_x(z).$$

Knowledge used. `durrett_ch5_stationary_measure_from_returns`;
`durrett_ch5_stationary_measure_uniqueness_recurrent`.

Solution. For each recurrent state a , the cycle trick defines a stationary measure μ_a with normalization $\mu_a(a) = 1$. Since the chain is irreducible and recurrent, the stationary measure is unique up to a constant multiple.

Thus μ_x and μ_y are multiples of one another. Since $\mu_y(y) = 1$, evaluating the proportionality at y gives

$$\mu_x = \mu_x(y)\mu_y.$$

Evaluating this identity at z yields

$$\mu_x(z) = \mu_x(y)\mu_y(z),$$

which is the desired formula.

Exercise 5.5.4

Question. Suppose p is irreducible and positive recurrent. Then

$$E_x T_y < \infty \quad \text{for all } x, y.$$

Knowledge used. [durrett_ch5_positive_recurrence_equivalences](#);
[durrett_ch5_positive_recurrence_return_time](#); [durrett_ch5_strong_markov_property](#).

Solution. Since the chain is irreducible and positive recurrent, every state is positive recurrent. In particular,

$$E_y T_y < \infty.$$

We show that this forces $E_x T_y < \infty$ for all x .

By irreducibility, choose a path from y to x with positive probability. Taking the first such path with minimal length, it avoids y until its endpoint x . Hence there are $K < \infty$ and $a > 0$ such that

$$P_y(X_K = x, T_y > K) \geq a.$$

If $E_x T_y = \infty$, then on the event above the return time to y would have infinite conditional mean after time K . By the strong Markov property,

$$E_y T_y \geq a(K + E_x T_y) = \infty,$$

contradicting positive recurrence of y . Therefore $E_x T_y < \infty$.

Exercise 5.5.5

Question. Suppose p is irreducible and has a stationary measure μ with

$$\sum_x \mu(x) = \infty.$$

Then p is not positive recurrent.

Knowledge used. [durrett_ch5_stationary_measure_uniqueness_recurrent](#);
[durrett_ch5_positive_recurrence_equivalences](#); [durrett_ch5_positive_recurrence_return_time](#).

Solution. Suppose, for contradiction, that p is positive recurrent. Since the chain is irreducible, positive recurrence is equivalent to the existence of a stationary distribution π .

For an irreducible positive recurrent chain, every stationary measure is a constant multiple of π . Hence there is a constant $c \geq 0$ such that

$$\mu(x) = c\pi(x) \quad \text{for all } x.$$

It follows that

$$\sum_x \mu(x) = c \sum_x \pi(x) = c < \infty,$$

contradicting the hypothesis that μ has infinite total mass. Therefore p is not positive recurrent.

Exercise 5.5.6

Question. Use Theorems 5.5.7 and 5.5.9 to show that for simple random walk:

- (i) the expected number of visits to k between successive visits to 0 is 1 for all k ;
- (ii) if we start from k , the expected number of visits to k before hitting 0 is $2k$.

Knowledge used. `durrett_ch5_stationary_measure_from_returns`;
`durrett_ch5_stationary_measure_uniqueness_recurrent`; `durrett_ch5_random_walk_chain`.

Solution. For simple symmetric random walk on $\mathbb{Z}$, the counting measure

$$\nu(j) \equiv 1$$

is stationary, because the transition matrix is doubly stochastic. The chain is irreducible and recurrent. By uniqueness of stationary measures up to constants, the cycle measure μ_0 from Theorem 5.5.7 must be a constant multiple of counting measure. Since $\mu_0(0) = 1$, that constant is 1. Therefore

$$\mu_0(k) = 1 \quad \text{for every } k \in \mathbb{Z}.$$

This says exactly that the expected number of visits to k between successive visits to 0 is 1.

Now start from $k > 0$ and count visits to k before hitting 0, including the visit at time 0. Each time the walk is at k , the trial ends successfully if the walk hits 0 before returning to k . The probability of this success is

$$P_k(T_0 < T_k).$$

To hit 0 before returning to k , the first step must be to $k - 1$, which has probability $1/2$, and then the gambler's ruin probability of hitting 0 before k starting from $k - 1$ is $1/k$. Hence

$$P_k(T_0 < T_k) = \frac{1}{2k}.$$

The number of visits to k before hitting 0 is geometric with this success probability, so its mean is

$$\frac{1}{P_k(T_0 < T_k)} = 2k.$$

By symmetry the analogous value for $k < 0$ is $2|k|$.

Exercise 5.5.7

Question. Let $\xi_1, \xi_2, \dots$ be i.i.d. with $P(\xi_m = k) = a_{k+1}$ for $k \geq -1$, let

$$S_n = x + \xi_1 + \dots + \xi_n, \quad x \geq 0,$$

and let

$$X_n = S_n + \left(\min_{m \leq n} S_m \right)^-.$$

Equation (5.1.1) shows that X_n has the same distribution as the $M/G/1$ queue starting from $X_0 = x$. Use this representation to conclude that if $\mu = \sum_k k a_k < 1$, then as $n \rightarrow \infty$,

$$\frac{1}{n} |\{m \leq n : X_{m-1} = 0, \xi_m = -1\}| \longrightarrow 1 - \mu \quad \text{a.s.}$$

and hence $\pi(0) = (1 - \mu)/a_0$ as proved earlier.

Knowledge used. [durrett_ch5_mg1_queue_chain](#); [durrett_ch5_mg1_stability_stationarity](#); [durrett_ch5_stationary_measure_definition](#).

Solution. The recursion for the queue is

$$X_m = (X_{m-1} + \xi_m)^+.$$

Since $\xi_m \geq -1$, reflection occurs exactly on the event

$$\{X_{m-1} = 0, \xi_m = -1\}.$$

Let

$$L_n = |\{m \leq n : X_{m-1} = 0, \xi_m = -1\}|.$$

Then the reflected-walk identity is

$$X_n = x + \sum_{m=1}^n \xi_m + L_n.$$

Since $E\xi_1 = \mu - 1 < 0$, the strong law of large numbers gives

$$\frac{1}{n} \sum_{m=1}^n \xi_m \longrightarrow \mu - 1.$$

The explicit reflection representation also gives $X_n/n \rightarrow 0$ a.s.: the walk S_n has negative linear drift, and subtracting its running minimum leaves only a sublinear reflected height. Dividing the identity above by n yields

$$\frac{L_n}{n} = \frac{X_n}{n} - \frac{x}{n} - \frac{1}{n} \sum_{m=1}^n \xi_m \longrightarrow 1 - \mu \quad \text{a.s.}$$

In stationarity, the event $\{X_{m-1} = 0, \xi_m = -1\}$ has probability $\pi(0)a_0$, because ξ_m is independent of the past and $P(\xi_m = -1) = a_0$. The long-run frequency therefore equals $\pi(0)a_0$. Comparing with the a.s. limit above gives

$$\pi(0)a_0 = 1 - \mu, \quad \text{so} \quad \pi(0) = \frac{1 - \mu}{a_0}.$$

Exercise 5.5.8

Question. $M/M/\infty$ queue. In this chain, introduced in Exercise 5.3.8,

$$X_{n+1} = \sum_{m=1}^{X_n} \xi_{n,m} + Y_{n+1},$$

where the $\xi_{n,m}$ are i.i.d. Bernoulli with mean p and Y_{n+1} is an independent Poisson mean λ random variable. Compute the distribution at time 1 when X_0 is Poisson(θ) and use the result to find the stationary distribution.

Knowledge used. [durrett_ch5_stationary_measure_definition](#); [durrett_ch5_foster_supermartingale_recurrence](#).

Solution. If $X_0 \sim \text{Poisson}(\theta)$, then thinning a Poisson random variable shows that the number of old conversations that survive one more minute is

$$\sum_{m=1}^{X_0} \xi_{0,m} \sim \text{Poisson}(p\theta).$$

This random variable is independent of the new arrivals $Y_1 \sim \text{Poisson}(\lambda)$. Therefore

$$X_1 \sim \text{Poisson}(p\theta + \lambda).$$

For stationarity we need $p\theta + \lambda = \theta$, so

$$\theta = \frac{\lambda}{1-p}.$$

Thus the stationary distribution is

$$\pi(j) = e^{-\lambda/(1-p)} \frac{(\lambda/(1-p))^j}{j!}, \quad j = 0, 1, 2, \dots$$

Exercise 5.5.9

Question. Let $X_n \geq 0$ be a Markov chain and suppose

$$E_x X_1 \leq x - \epsilon \quad \text{for } x > K,$$

where $\epsilon > 0$. Let

$$Y_n = X_n + n\epsilon, \quad \tau = \inf\{n : X_n \leq K\}.$$

Then $Y_{n \wedge \tau}$ is a positive supermartingale and the optional stopping theorem implies $E_x \tau \leq x/\epsilon$. With assumptions about the behavior of the chain starting from points $x \leq K$, this leads to conditions for positive recurrence.

Knowledge used. [durrett_ch5_foster_supermartingale_recurrence](#); [durrett_ch5_positive_recurrence_equivalences](#); [durrett_ch5_hitting_return_decomposition](#).

Solution. For $n < \tau$, we have $X_n > K$, so

$$E(X_{n+1} \mid \mathcal{F}_n) \leq X_n - \epsilon.$$

Hence

$$E(Y_{n+1} \mid \mathcal{F}_n) = E(X_{n+1} \mid \mathcal{F}_n) + (n+1)\epsilon \leq X_n + n\epsilon = Y_n.$$

After τ the stopped process is constant, so $Y_{n \wedge \tau}$ is a nonnegative supermartingale.

By optional stopping,

$$E_x Y_{n \wedge \tau} \leq Y_0 = x.$$

Since $X_{n \wedge \tau} \geq 0$,

$$\epsilon E_x(n \wedge \tau) \leq x.$$

Letting $n \rightarrow \infty$ and using monotone convergence gives

$$E_x \tau \leq \frac{x}{\epsilon}.$$

Let $C = \{x : x \leq K\}$. If C is finite and the chain is irreducible on a countable state space, this drift estimate is the key Foster condition for positive recurrence. Indeed, from every outside state the expected time to hit C is finite. Starting from the finite set C , irreducibility gives a fixed state $a \in C$ that can be reached with positive probability in finitely many steps; failed attempts return to C in finite mean time by the drift estimate. A geometric-trials argument then gives $E_a T_a < \infty$. Thus a is positive recurrent, and by irreducibility all states are positive recurrent.

Exercise 5.6.1

Question. Suppose $S = \{0, 1\}$ and

$$p = \begin{pmatrix} 1 - \alpha & \alpha \\ \beta & 1 - \beta \end{pmatrix}.$$

Use induction to show that

$$P_\mu(X_n = 0) = \frac{\beta}{\alpha + \beta} + (1 - \alpha - \beta)^n \left\{ \mu(0) - \frac{\beta}{\alpha + \beta} \right\}.$$

Knowledge used. [durrett_ch5_stationary_measure_definition](#);
[durrett_ch5_aperiodic_convergence_theorem](#).

Solution. Let

$$q_n = P_\mu(X_n = 0).$$

Then

$$q_{n+1} = q_n(1 - \alpha) + (1 - q_n)\beta = \beta + (1 - \alpha - \beta)q_n.$$

The fixed point of this affine recursion is

$$q_* = \frac{\beta}{\alpha + \beta},$$

since $q_* = \beta + (1 - \alpha - \beta)q_*$. Therefore

$$q_{n+1} - q_* = (1 - \alpha - \beta)(q_n - q_*).$$

Iterating from $q_0 = \mu(0)$ gives

$$q_n - q_* = (1 - \alpha - \beta)^n (\mu(0) - q_*),$$

which is the desired formula.

Exercise 5.6.2

Question. Show that if S is finite and p is irreducible and aperiodic, then there is an m such that

$$p^m(x, y) > 0 \quad \text{for all } x, y.$$

Knowledge used. [durrett_ch5_period_definition](#);
[durrett_ch5_aperiodic_convergence_theorem](#); [durrett_ch5_finite_closed_recurrent_class](#).

Solution. Because the chain is irreducible, for each pair x, y there is a $k(x, y)$ such that

$$p^{k(x, y)}(x, y) > 0.$$

Because the chain is aperiodic, Lemma 5.6.5 from the textbook says that for each y there is an $M(y)$ such that

$$p^\ell(y, y) > 0 \quad \text{for all } \ell \geq M(y).$$

The state space is finite, so choose

$$m \geq \max_{x, y} \{k(x, y) + M(y)\}.$$

Then for every x, y ,

$$p^m(x, y) \geq p^{k(x, y)}(x, y) p^{m-k(x, y)}(y, y) > 0.$$

Exercise 5.6.3

Question. Show that if S is finite, p is irreducible and aperiodic, and T is the coupling time defined in the proof of Theorem 5.6.6, then

$$P(T > n) \leq Cr^n$$

for some $r < 1$ and $C < \infty$. Thus convergence to equilibrium occurs exponentially rapidly in this case. Hint: first consider the case in which $p(x, y) > 0$ for all x, y and reduce the general case to this one by looking at a power of p .

Knowledge used. [durrett_ch5_aperiodic_convergence_theorem](#); [durrett_ch5_coupling_for_finite_chains](#); [durrett_ch5_period_definition](#).

Solution. First suppose $p(x, y) > 0$ for all x, y . Since S is finite, choose a state a and set

$$\eta = \min_{x \in S} p(x, a) > 0.$$

For the independent-coordinate coupling used in Theorem 5.6.6, from any current pair (x, y) the probability that both coordinates jump to a in the next step is at least η^2 . Hence

$$P(T > n + 1 \mid T > n) \leq 1 - \eta^2,$$

and therefore

$$P(T > n) \leq (1 - \eta^2)^n.$$

In the general finite irreducible aperiodic case, Exercise 5.6.2 gives an integer m such that

$$p^m(x, y) > 0 \quad \text{for all } x, y.$$

Apply the previous argument to the m -step skeleton chain. There is $\eta > 0$ such that, at each block of m steps, the coupled coordinates meet with probability at least η^2 . Thus

$$P(T > km) \leq (1 - \eta^2)^k.$$

For arbitrary n , take $k = \lfloor n/m \rfloor$ and write this as

$$P(T > n) \leq Cr^n$$

for suitable constants $C < \infty$ and $r = (1 - \eta^2)^{1/m} < 1$.

Exercise 5.6.4

Question. For any transition matrix p , define

$$\alpha_n = \sup_{i,j} \frac{1}{2} \sum_k |p^n(i, k) - p^n(j, k)|.$$

The factor $1/2$ is there because for any i, j we can define random variables X and Y such that $P(X = k) = p^n(i, k)$, $P(Y = k) = p^n(j, k)$, and

$$P(X \neq Y) = \frac{1}{2} \sum_k |p^n(i, k) - p^n(j, k)|.$$

Show that

$$\alpha_{m+n} \leq \alpha_n \alpha_m.$$

Knowledge used. [durrett_ch5_coupling_for_finite_chains](#); [durrett_ch5_aperiodic_convergence_theorem](#).

Solution. We use the coupling interpretation. Start two copies of the chain from i and j . Couple their positions after m steps as efficiently as possible, so that

$$P(X_m \neq Y_m) \leq \alpha_m.$$

If $X_m = Y_m$, then use the same future randomness for the two chains, so they remain together. If $X_m \neq Y_m$, couple the next n steps as efficiently as possible; conditional on the two different states at time m , the chance they fail to couple after another n steps is at most α_n .

Therefore

$$P(X_{m+n} \neq Y_{m+n}) \leq \alpha_m \alpha_n.$$

Taking the best possible coupling interpretation of total variation and then the supremum over i, j gives

$$\alpha_{m+n} \leq \alpha_m \alpha_n.$$

Exercise 5.6.5

Question. Strong law for additive functionals. Suppose p is irreducible and has stationary distribution π . Let f satisfy

$$\sum_y |f(y)| \pi(y) < \infty.$$

Let T_x^k be the time of the k th return to x .

(i) Show that

$$V_k^f = f(X(T_x^k)) + \cdots + f(X(T_x^{k+1} - 1)), \quad k \geq 1,$$

are i.i.d. and have $E|V_k^f| < \infty$.

(ii) Let $K_n = \inf\{k : T_x^k \geq n\}$ and show that

$$\frac{1}{n} \sum_{m=1}^{K_n} V_m^f \longrightarrow \frac{EV_1^f}{E_x T_x^1} = \sum_y f(y) \pi(y) \quad P_\mu\text{-a.s.}$$

(iii) Show that

$$\frac{\max_{1 \leq m \leq n} |V_m^f|}{n} \rightarrow 0$$

and conclude that

$$\frac{1}{n} \sum_{m=1}^n f(X_m) \longrightarrow \sum_y f(y) \pi(y) \quad P_\mu\text{-a.s.}$$

for any initial distribution μ .

Knowledge used. `durrett_ch5_additive_functional_lln`;
`durrett_ch5_stationary_measure_from_returns`; `durrett_ch5_positive_recurrence_return_time`;
`durrett_ch5_strong_markov_property`.

Solution. The return times to x split the path into regenerative cycles. By the strong Markov property, the cycle rewards

$$V_k^f = \sum_{r=T_x^k}^{T_x^{k+1}-1} f(X_r), \quad k \geq 1,$$

are i.i.d.

The cycle occupation measure satisfies

$$\mu_x(y) = E_x \sum_{r=0}^{T_x^1-1} \mathbf{1}_{\{X_r=y\}}.$$

Since π is stationary and the chain is irreducible positive recurrent,

$$\mu_x(y) = \frac{\pi(y)}{\pi(x)}, \quad E_x T_x^1 = \frac{1}{\pi(x)}.$$

Therefore

$$E|V_1^f| \leq \sum_y |f(y)| \mu_x(y) = \frac{1}{\pi(x)} \sum_y |f(y)| \pi(y) < \infty,$$

and

$$EV_1^f = \frac{1}{\pi(x)} \sum_y f(y) \pi(y).$$

By the strong law of large numbers applied to the iid cycle lengths and cycle rewards,

$$\frac{T_x^k}{k} \rightarrow E_x T_x^1, \quad \frac{1}{k} \sum_{m=1}^k V_m^f \rightarrow EV_1^f.$$

Since $K_n/n \rightarrow 1/E_x T_x^1$, random indexing gives

$$\frac{1}{n} \sum_{m=1}^{K_n} V_m^f = \frac{K_n}{n} \cdot \frac{1}{K_n} \sum_{m=1}^{K_n} V_m^f \rightarrow \frac{EV_1^f}{E_x T_x^1} = \sum_y f(y) \pi(y).$$

If $Z_m = V_m^{|f|}$, then the Z_m are iid and integrable. A standard dyadic Borel-Cantelli argument for iid integrable variables gives

$$\frac{\max_{1 \leq m \leq n} Z_m}{n} \rightarrow 0 \quad \text{a.s.}$$

The difference between the ordinary partial sum $\sum_{m=1}^n f(X_m)$ and the completed-cycle sum is bounded in absolute value by the reward in at most one incomplete cycle, plus an initial finite entrance segment if the chain does not start at x . Dividing by n , these errors vanish a.s. Hence

$$\frac{1}{n} \sum_{m=1}^n f(X_m) \rightarrow \sum_y f(y) \pi(y) \quad P_\mu\text{-a.s.}$$

Exercise 5.6.6

Question. Central limit theorem for additive functionals. Suppose in addition to the conditions in Exercise 5.6.5 that

$$\sum_y f(y)\pi(y) = 0, \quad E_x(V_k^{|f|})^2 < \infty.$$

(i) Use the random-index central limit theorem to conclude that for any initial distribution μ ,

$$\frac{1}{\sqrt{n}} \sum_{m=1}^{K_n} V_m^f \Rightarrow c\chi \quad \text{under } P_\mu,$$

where χ is standard normal.

(ii) Show that

$$\frac{\max_{1 \leq m \leq n} V_m^{|f|}}{\sqrt{n}} \rightarrow 0 \quad \text{in probability}$$

and conclude that

$$\frac{1}{\sqrt{n}} \sum_{m=1}^n f(X_m) \Rightarrow c\chi \quad \text{under } P_\mu.$$

Knowledge used. [durrett_ch5_additive_functional_lln](#); [durrett_ch3_converging_together_slutsky](#); [durrett_ch3_lindeberg_feller_clt](#).

Solution. From Exercise 5.6.5, the cycle rewards V_m^f are iid. The assumption $\sum_y f(y)\pi(y) = 0$ implies

$$EV_1^f = 0.$$

The square-integrability assumption gives

$$\sigma^2 = \text{Var}(V_1^f) < \infty.$$

Also $K_n/n \rightarrow 1/E_x T_x^1$ in probability. The random-index central limit theorem therefore gives

$$\frac{1}{\sqrt{n}} \sum_{m=1}^{K_n} V_m^f \Rightarrow N\left(0, \frac{\sigma^2}{E_x T_x^1}\right).$$

Thus $c = \sigma / \sqrt{E_x T_x^1}$.

Let $Z_m = V_m^{|f|}$. Since $E Z_1^2 < \infty$,

$$P\left(\max_{1 \leq m \leq n} Z_m > \varepsilon \sqrt{n}\right) \leq nP(Z_1 > \varepsilon \sqrt{n}).$$

The right side tends to 0 by the usual tail consequence of square-integrability; equivalently, split into dyadic blocks and use Borel-Cantelli. Hence

$$\frac{\max_{1 \leq m \leq n} V_m^{|f|}}{\sqrt{n}} \rightarrow 0 \quad \text{in probability.}$$

The difference between the completed-cycle reward and the ordinary partial sum is bounded by the reward in one incomplete cycle, plus a finite initial segment when the initial distribution is not concentrated at x . Dividing by $\sqrt{n}$, this difference goes to 0 in probability. Slutsky's theorem gives

$$\frac{1}{\sqrt{n}} \sum_{m=1}^n f(X_m) \Rightarrow c\chi.$$

Exercise 5.6.7

Question. Ratio limit theorems. Theorem 5.6.1 does not say much in the null recurrent case. To get a more informative limit theorem, suppose that y is recurrent and m is the stationary measure on $C_y = \{z : \rho_{yz} > 0\}$, unique up to constant multiples. Let

$$N_n(z) = |\{m \leq n : X_m = z\}|.$$

Break up the path at successive returns to y and show that

$$\frac{N_n(z)}{N_n(y)} \longrightarrow \frac{m(z)}{m(y)} \quad P_x\text{-a.s.}$$

for all $x, z \in C_y$.

Knowledge used. [durrett_ch5_renewal_limit_return_probabilities](#);
[durrett_ch5_stationary_measure_from_returns](#);
[durrett_ch5_stationary_measure_uniqueness_recurrent](#).

Solution. Normalize the stationary measure so that $m(y) = 1$. With this normalization, $m = \mu_y$, the cycle occupation measure generated by successive returns to y .

Let R_k be the k th return time to y . In the k th cycle, define

$$Z_k(z) = \sum_{r=R_k}^{R_{k+1}-1} \mathbf{1}_{\{X_r=z\}}.$$

By the strong Markov property, the $Z_k(z)$ are iid for $k \geq 1$, and

$$EZ_1(z) = \mu_y(z) = m(z).$$

Also each complete cycle contains exactly one visit to y , namely at its start.

By the strong law of large numbers,

$$\frac{1}{K} \sum_{k=1}^K Z_k(z) \rightarrow m(z).$$

Since $N_n(y) \rightarrow \infty$ a.s. on the recurrent class and the unfinished last cycle contributes only a negligible error when divided by $N_n(y)$, we get

$$\frac{N_n(z)}{N_n(y)} \rightarrow m(z).$$

Restoring the original normalization gives

$$\frac{N_n(z)}{N_n(y)} \rightarrow \frac{m(z)}{m(y)}.$$

If the chain starts from $x \in C_y$ rather than from y , the finite initial segment before the first hit of y is negligible in the same ratio.

Exercise 5.6.8

Question. We got (5.6.1) from Theorem 5.6.1 by taking expected values. This does not work for the ratio in the previous exercise, so we need another approach. Suppose $z \neq y$.

(i) Let

$$\bar{p}_n(x, z) = P_x(X_n = z, T_y > n)$$

and decompose $p^m(x, z)$ according to the value of

$$J = \sup\{j \in [1, m) : X_j = y\}$$

to get

$$\sum_{m=1}^n p^m(x, z) = \sum_{m=1}^n \bar{p}_m(x, z) + \sum_{j=1}^{n-1} p^j(x, y) \sum_{k=1}^{n-j} \bar{p}_k(y, z).$$

(ii) Show that

$$\frac{\sum_{m=1}^n p^m(x, z)}{\sum_{m=1}^n p^m(x, y)} \longrightarrow \frac{m(z)}{m(y)}.$$

Knowledge used. [durrett_ch5_renewal_limit_return_probabilities](#);
[durrett_ch5_hitting_return_decomposition](#); [durrett_ch5_stationary_measure_from_returns](#).

Solution. For part (i), either the path has no visit to y before time m , in which case its contribution to $P_x(X_m = z)$ is $\bar{p}_m(x, z)$, or it has a last visit to y before time m . If this last visit occurs at time j , then the segment after time j starts from y , ends at z after $k = m - j$ steps, and avoids y in between. By the Markov property, that contribution is

$$p^j(x, y) \bar{p}_k(y, z).$$

Summing over $m \leq n$ and writing $k = m - j$ gives the displayed formula.

For part (ii), normalize m so that $m(y) = 1$. Then the cycle formula for the stationary measure gives

$$\sum_{k=1}^{\infty} \bar{p}_k(y, z) = \mu_y(z) = m(z),$$

because $z \neq y$. Hence

$$B_N(z) := \sum_{k=1}^N \bar{p}_k(y, z) \longrightarrow m(z).$$

The first term in the decomposition,

$$\sum_{m=1}^n \bar{p}_m(x, z),$$

is bounded as $n \rightarrow \infty$ by the expected number of visits to z before hitting y . Meanwhile

$$\sum_{m=1}^n p^m(x, y) \rightarrow \infty$$

because y is recurrent and $x \in C_y$.

Using the decomposition from part (i),

$$\sum_{m=1}^n p^m(x, z) = O(1) + \sum_{j=1}^{n-1} p^j(x, y) B_{n-j}(z).$$

This is a weighted average of $B_{n-j}(z)$ with nonnegative weights $p^j(x, y)$, plus a negligible bounded term. Since $B_N(z) \rightarrow m(z)$ and the total weight $\sum_{j=1}^{n-1} p^j(x, y)$ diverges, the Toeplitz averaging lemma gives

$$\frac{\sum_{m=1}^n p^m(x, z)}{\sum_{m=1}^n p^m(x, y)} \rightarrow m(z).$$

Undoing the normalization $m(y) = 1$ yields

$$\frac{\sum_{m=1}^n p^m(x, z)}{\sum_{m=1}^n p^m(x, y)} \rightarrow \frac{m(z)}{m(y)}.$$

Exercise 5.8.1

Question. Show that $\bar{X}_n$ is recurrent if and only if

$$\sum_{n=1}^{\infty} \bar{p}^n(\alpha, \alpha) = \infty.$$

Knowledge used. [durrett_ch5_harris_recurrence](#); [durrett_ch5_recurrence_visit_criterion](#); [durrett_ch5_small_set_split_chain](#).

Solution. For the split chain $\bar{X}_n$, the artificial point α is an atom. Let

$$R = \inf\{n \geq 1 : \bar{X}_n = \alpha\}.$$

By definition, the Harris chain is recurrent when $P_\alpha(R < \infty) = 1$. Starting from α , successive returns to α regenerate the process. Hence, exactly as in the countable-state return criterion,

$$E_\alpha \sum_{n=1}^{\infty} \mathbf{1}_{\{\bar{X}_n = \alpha\}} = \sum_{n=1}^{\infty} \bar{p}^n(\alpha, \alpha) = \begin{cases} \infty, & P_\alpha(R < \infty) = 1, \\ \frac{P_\alpha(R < \infty)}{1 - P_\alpha(R < \infty)}, & P_\alpha(R < \infty) < 1. \end{cases}$$

Thus $\bar{X}_n$ is recurrent if and only if the displayed series diverges.

Exercise 5.8.2

Question. If X_n is a recurrent Harris chain on a countable state space, then S can only have one irreducible set of recurrent states but may have a nonempty set of transient states. For a concrete example, consider a branching process in which the probability of no children $p_0 > 0$ and set $A = B = \{0\}$.

Knowledge used. [durrett_ch5_harris_recurrence](#); [durrett_ch5_decomposition_theorem](#); [durrett_ch5_branching_process_chain](#).

Solution. For a recurrent Harris chain, the split-chain atom α is recurrent. Theorem 5.8.8 says that every set with positive λ -measure is visited infinitely often from α . In a countable state space, states in recurrent classes that are visible from the Harris set must therefore communicate through the

same recurrent split-chain structure. Consequently there can be at most one irreducible recurrent class. States outside that class may still be transient, because Harris recurrence only forces repeated visits to sets seen by the recurrent atom, not recurrence of every starting state.

For the branching-process example, state 0 is absorbing. If $p_0 > 0$, then from any state $k \geq 1$ there is positive probability p_0^k that all individuals have no offspring in the next generation, so the chain can hit 0. Taking $A = B = \{0\}$ gives the Harris minorization, and the split-chain atom is recurrent because once the process reaches 0 it stays there.

The unique recurrent class is $\{0\}$. The states $k \geq 1$ are not in that recurrent class; when extinction occurs they are never visited again. Thus the recurrent Harris chain may have transient states.

Exercise 5.8.3

Question. Use Exercise 5.8.1 and imitate the proof of Theorem 5.5.10 to show that a Harris chain with a stationary distribution must be recurrent.

Knowledge used. `durrett_ch5_harris_recurrence`; `durrett_ch5_harris_stationary_measure`; `durrett_ch5_stationary_measure_definition`.

Solution. Let π be a stationary distribution for the original Harris chain. Form the split-chain measure

$$\bar{\pi} = \pi \bar{p}.$$

By Lemma 5.8.10, $\bar{\pi}$ is stationary for $\bar{p}$, and

$$\bar{\pi}(\alpha) = \epsilon\pi(A) > 0$$

for the Harris set A .

Stationarity gives $\bar{\pi}\bar{p}^n = \bar{\pi}$ for all n , so

$$\sum_{n=1}^{\infty} \bar{\pi}(\alpha) = \infty.$$

Equivalently,

$$\int \bar{\pi}(dx) \sum_{n=1}^{\infty} \bar{p}^n(x, \alpha) = \infty.$$

If α were transient, then the expected number of visits to α starting from any state would be bounded by a finite multiple of the probability of ever hitting α , because after each hit the future returns are governed by a geometric number of trials with success probability $P_\alpha(R < \infty) < 1$. The integral above would then be finite, a contradiction. Hence

$$\sum_{n=1}^{\infty} \bar{p}^n(\alpha, \alpha) = \infty.$$

By Exercise 5.8.1, the Harris chain is recurrent.

Exercise 5.8.4

Question. Suppose X_n is a recurrent Harris chain. Show that if (A', B') is another pair satisfying the conditions of the definition, then Theorem 5.8.8 implies

$$P_\alpha(\bar{X}_n \in A' \text{ i.o.}) = 1,$$

so the recurrence or transience does not depend on the choice of (A, B) .

Knowledge used. `durrett_ch5_harris_recurrence`; `durrett_ch5_small_set_split_chain`; `durrett_ch5_harris_stationary_measure`.

Solution. Let λ be the irreducibility measure associated with the split chain built from (A, B) :

$$\lambda(C) = \sum_{n=1}^{\infty} 2^{-n} \bar{p}^n(\alpha, C).$$

Because (A', B') also satisfies the Harris definition, every state in S has positive probability of eventually hitting A' . Starting from α , the split chain first redistributes into S according to the measure ρ associated with (A, B) , and then follows the original chain. Therefore

$$P_\alpha(\tau_{A'} < \infty) > 0.$$

This implies $\lambda(A') > 0$.

Since the chain is recurrent, Theorem 5.8.8 applies to every set of positive λ -measure. Hence

$$P_\alpha(\bar{X}_n \in A' \text{ i.o.}) = 1.$$

Thus the atom built from (A, B) visits the other Harris set A' infinitely often. Reversing the roles of the two pairs gives the same conclusion in the other split chain. Consequently the recurrence or transience classification is intrinsic and does not depend on the particular choice of (A, B) .

Exercise 5.8.5

Question. In the GI/G/1 queue, the waiting time W_n and the random walk

$$S_n = X_0 + \xi_1 + \cdots + \xi_n$$

agree until

$$N = \inf\{n : S_n < 0\},$$

and at this time $W_N = 0$. Use this observation as in Example 5.3.7 to show that Example 5.8.3 is transient when $E\xi_n > 0$, recurrent when $E\xi_n \leq 0$, and has a stationary distribution when $E\xi_n < 0$.

Knowledge used. `durrett_ch5_gig1_queue_storage_model`; `durrett_ch5_harris_recurrence`; `durrett_ch5_foster_supermartingale_recurrence`; `durrett_ch5_harris_stationary_measure`.

Solution. The recursion is

$$W_{n+1} = (W_n + \xi_{n+1})^+.$$

Until the ordinary random walk

$$S_n = X_0 + \xi_1 + \cdots + \xi_n$$

first goes below 0, the reflection has not acted, so $W_n = S_n$. At the first time $N = \inf\{n : S_n < 0\}$, the reflected process is reset to 0.

If $E\xi_1 > 0$, then $S_n \rightarrow \infty$ with positive probability when X_0 is large enough. On that event the random walk never drops below 0, so the queue never returns to 0. Hence the Harris chain is transient.

If $E\xi_1 \leq 0$, the one-dimensional random walk is oscillatory or drifts to $-\infty$, and therefore it eventually drops below 0 a.s. from every finite initial workload. Thus W_n hits 0 a.s. With $A = B = \{0\}$ as in Example 5.8.3, this gives recurrence of the Harris chain.

If $E\xi_1 < 0$, the drift toward 0 is strict. The same supermartingale argument used for the M/G/1 queue gives

$$E_x N < \infty$$

for every x . Thus the return time to the Harris atom has finite mean, so the recurrent Harris chain is positive recurrent and has a stationary distribution.

Exercise 5.8.6

Question. Let

$$m_n = \min(S_0, S_1, \dots, S_n),$$

where S_n is the random walk defined previously.

(i) Show that

$$S_n - m_n \stackrel{d}{=} W_n.$$

(ii) Let $\xi'_m = \xi_{n+1-m}$ for $1 \leq m \leq n$. Show that

$$S_n - m_n = \max(S'_0, S'_1, \dots, S'_n).$$

(iii) Conclude that as $n \rightarrow \infty$,

$$W_n \Rightarrow M \equiv \max(S'_0, S'_1, S'_2, \dots).$$

Knowledge used. [durrett_ch5_gig1_queue_storage_model](#); [durrett_ch5_random_walk_chain](#); [durrett_ch3_weak_convergence_definition](#).

Solution. Assume $W_0 = S_0 = 0$. The reflected recursion has the explicit Skorokhod-reflection form

$$W_n = S_n - \min_{0 \leq j \leq n} S_j$$

as can be checked by induction. Indeed, if the running minimum does not decrease at time $n+1$, then both sides add ξ_{n+1} ; if the running minimum decreases, then the right side becomes 0, matching the positive-part recursion. Hence

$$W_n = S_n - m_n.$$

This proves the distributional statement, and in this construction the two sides are actually equal.

For part (ii),

$$S_n - m_n = \max_{0 \leq j \leq n} (S_n - S_j) = \max_{0 \leq j \leq n} (\xi_{j+1} + \dots + \xi_n).$$

If $\xi'_m = \xi_{n+1-m}$ and $S'_k = \xi'_1 + \dots + \xi'_k$, then the sums $\xi_{j+1} + \dots + \xi_n$ are exactly S'_{n-j} for $0 \leq j \leq n$. Therefore

$$S_n - m_n = \max_{0 \leq k \leq n} S'_k.$$

If $E\xi_1 < 0$, then the reversed increments have the same distribution as the original increments and the random walk S'_k drifts to $-\infty$. Hence

$$M = \max_{k \geq 0} S'_k < \infty \quad \text{a.s.}$$

The finite maxima $\max_{0 \leq k \leq n} S'_k$ increase a.s. to M . Since W_n has the same distribution as this finite maximum,

$$W_n \Rightarrow M.$$

Exercise 5.8.7

Question. In the discrete O.U. process, X_{n+1} is normal with mean θX_n and variance 1. What happens to the recurrence and transience if instead Y_{n+1} is normal with mean 0 and variance $\beta^2 |Y_n|$?

Knowledge used. [durrett_ch5_harris_recurrence](#); [durrett_ch5_harris_convergence_theorem](#); [durrett_ch5_foster_supermartingale_recurrence](#).

Solution. If $Y_n = 0$, then the conditional variance is 0, so 0 is absorbing. Starting from $Y_0 \neq 0$, the transition distribution is continuous and puts probability zero on the single point 0. Thus $\{0\}$ is a separate absorbing recurrent class, while the nonzero states form the interesting continuous class.

For $Y_n \neq 0$, write $R_n = |Y_n|$ and $L_n = \log R_n$. If Z_{n+1} is a standard normal random variable, then

$$Y_{n+1} = \beta \sqrt{|Y_n|} Z_{n+1},$$

so

$$L_{n+1} = \log \beta + \frac{1}{2} L_n + \log |Z_{n+1}|.$$

This is an autoregressive recursion on the log-scale with coefficient 1/2. Since $E|\log |Z|| < \infty$, the recursion is stable and has a proper stationary distribution:

$$L_\infty \stackrel{d}{=} \sum_{j=0}^{\infty} 2^{-j} (\log \beta + \log |Z_{j+1}|).$$

Thus $|Y_n|$ neither escapes to infinity nor collapses to zero from a nonzero start; it returns repeatedly to bounded log-scale sets.

Consequently, for every $\beta > 0$, the chain restricted to $\mathbb{R} \setminus \{0\}$ is recurrent in the Harris sense and in fact has a stationary distribution. The full chain is not irreducible because 0 is absorbing and cannot be reached from nonzero states, but its recurrence structure is clear: $\{0\}$ is an absorbing recurrent class, and the nonzero Harris class is positive recurrent.